Bihar Engineering University, Patna
End Semester Examination - 2023

Course: B.Tech.
Code: 105402
Semester-IV
Subject: Design and Analysis of Algorithms
Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (Any seven question only): **[2 x 7 = 14]**

(a) The fractional Knapsack problem can be solved by using
(i) Greedy method (ii) Divided and conquer method
(iii) Dynamic programming (iv) None of these

(b) BFS on a graph G = (V,E) has running time
(i) O (| V | + | E |) (ii) O (| V |)
(iii) O (| E |) (iv) None of the above

(c) The minimum number of colors needed to color a graph having n > 3 vertices and 2 edges is
(i) 2 (ii) 3
(iii) 4 (iv) 1

(d) Complexity the recurrence relation $T(n) = 8T\left(\frac{n}{2}\right) + n^2$ is
(i) O (n) (ii) O (n^2)
(iii) O ($\log_2 n$) (iv) O (n^3)

(e) Travelling salesman problem belongs to
(i) P class
(ii) NP class
(iii) NP- hard
(iv) NP- complete class

(f) Kruskal's algorithm uses _____ and Prism' algorithm uses _____ in determining the MST
(i) Edges, vertex (ii) vertex, edges
(iii) Edges, edges (iv) Vertex, vertex

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(g) Level order traversal of a rooted tree can be done by starting from root and performing
(i) Depth first search (ii) Breadth first search
(iii) Pre-order traversal (iv) In-order traversal

(h) An algorithm is made up of two independent time complexities $f(n)$ and g (n). Then the complexities of the algorithm is in order of
(i) $f(n)$x g(n) (ii) max($f(n)$).g(n)
(iii) min ($f(n)$.g(n) (iv) $f(n)$+g(n)

(i) Which of the following standard algorithms is not a greedy alogorithm?
(i) Dijkstra's shortest path algorithm (ii) Kruskal algorithm
(iii) Bellmen ford shortest path algorithm (iv) Prim's algorithm

(j) The node removal of which makes a graph disconnected is called
(i) Pendant vertex
(ii) Bridge
(iii) Articulation point
(iv) Coloured vertex

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Q.2 (a) Discuss the average, worst best time complexity of the algorithm. Give suitable examples. [7]

(b) Write the algorithm for quick-sort and find its complexity. [7]

Q.3 (a) Construct the Huffman coding tree for the text of characters with given frequencies : [7]

Characters	T	I	V	K	L	E	Q	Z	P	R
Frequencies	43	38	16	8	56	12	41	13	22	6

(b) State the general Knapsack problem. Write a greedy algorithm for this problem and derive its time complexity. [7]

Q.4 (a) State master's theorem and find the time complexity for the following recurrence: [6]
$T(n) = 2T(n^{1/2}) + \log n$

(b) What is negative weight-cycle? Write Bellman- Ford algorithm to find single source shortest distance of a directed graph. [8]

Q.5 (a) Find the minimum number of operation required for the following matix chain multiplication using dynamic programming [7]
A (10 x 20) * B (20 x 50) * C (50 x 1) * D (1 x 100)

(b) Write Knuth-Morirs-Pratt algorithm for string matching problem. [7]

Q.6 (a) Write an algorithm to find a minimum spanning tree (MST) for an undirected graph. Estimate the time complexity of your algorithm. [7]

(b) Using greedy strategy. Schedule the following jobs within deadline so as to maximize the profit. Deadline and profits are mentioned as follow: [7]

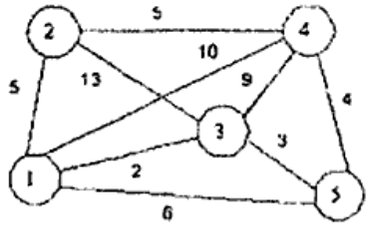
Job i	1	2	3	4
Deadline d	3	2	3	1
Profit g	9	7	7	2

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Q.7 (a) Write an algorithm for n-queen's problem find its time- complexity and explain the algorithm using an example. [7]

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(b) Solve the single source shortest path problem for the following graph considering '1' as the source vertex using Dijkstra's algorithm. [7]



Q.8 (a) Define the classes P and NP. [2]

(b) Discuss what you mean by polynomial reduction.. [2]

(c) Discuss diagrammatically the relation among P class, NP class, NP hard and NP complete. [2]

(d) Describe Clique Decision Problem (CDP) [2]

(e) Explain the max-flow min-cut theorem with an example. [6]

Q.9 Write short notes on *any two* of the following: [7x2=14]

(a) Asymptotic notations

(b) Heap creation technique

(c) Strassen's matrix multiplication

(d) Divide-and-Conquer vs Dynamic programming.

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